2022 VCE Specialist Mathematics 2 (NHT) external assessment report

Specific information

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

The statistics in this report may be subject to rounding resulting in a total more or less than 100 per cent.

Section A – Multiple-choice questions

Question	Answer
1	E
2	D
3	C
4	D
5	B
6	E
7	A
8	B
9	C
10	E
11	A
12	C
13	C
14	D
15	B
16	B
17	A
18	D
19	E
20	E

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Section B

Question 1ai.

$$f(x)=x+\frac{1}{2}\times\frac{1}{x-1}+\frac{1}{2}\times\frac{1}{x+1}$$

Question 1aii.

$y=x,\ x=1,\ x=-1$

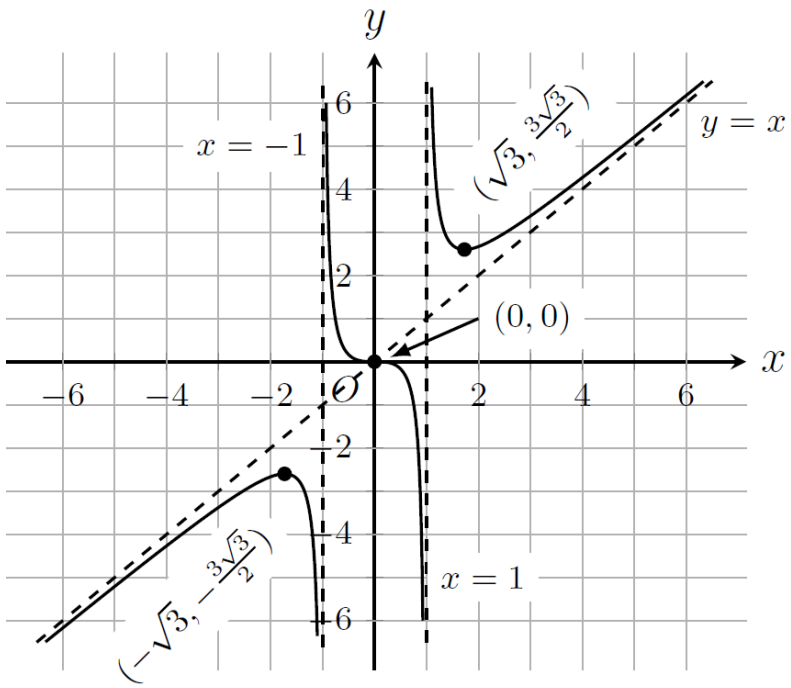
Question 1bi.

$$f'(x)=\frac{x^2\left(x^2-3\right)}{\left(x^2-1\right)^2}$$

Question 1bii.

$\left(-\sqrt{3},-\frac{3\sqrt{3}}{2}\right)$ local max, $(0,0)$ (stationary) point of inflection, $\left(\sqrt{3},\frac{3\sqrt{3}}{2}\right)$ local min

Question 1c.



Question 1di.

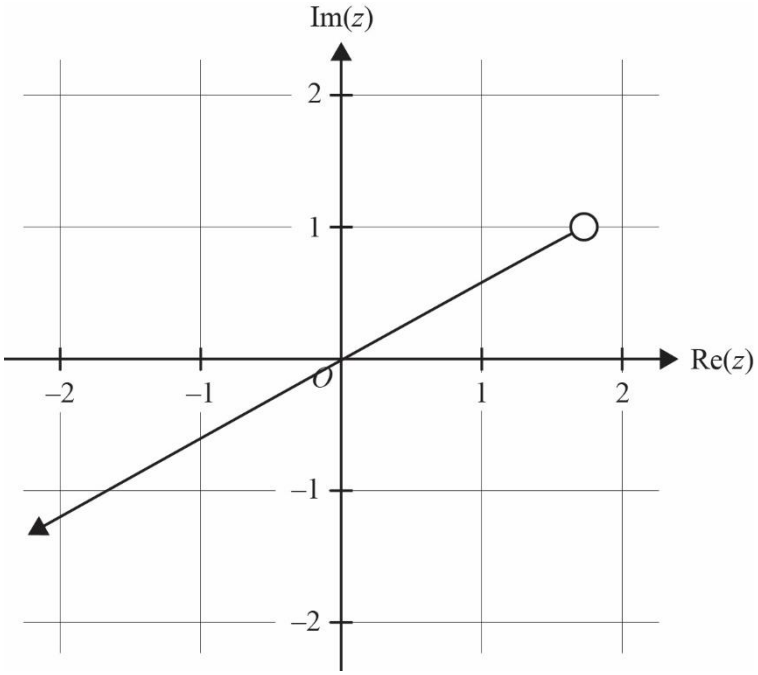
$$\pi\int_2^4\frac{x^6}{\left(x^2-1\right)^2}\,dx$$

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Question 1dii.

74.1

Question 2a.



Question 2b.

$-\frac{5\pi}{6}$

Question 2c.

$|z| = 2$ or other form, such as $\sqrt{z\,\overline{z}} = 2$

Question 2d.

$z_1 = -\sqrt{3} + i$, $z_2 = -2i$ or polar form $2\text{cis}\left(\frac{5\pi}{6}\right)$, $2\text{cis}\left(-\frac{\pi}{2}\right)$

Question 2e.

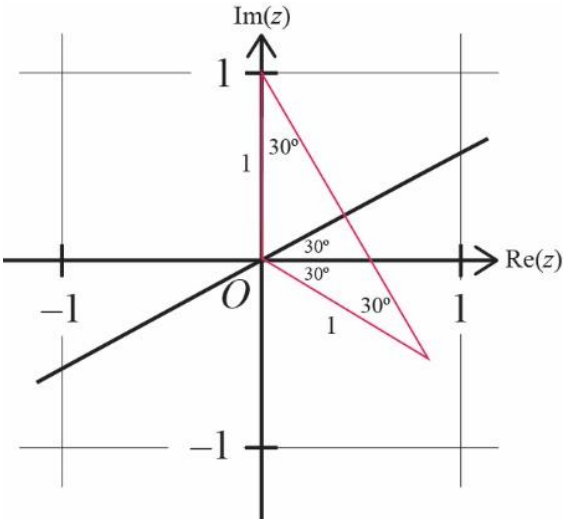
$f(z) = \left(z - \left(\sqrt{3} + i\right)\right)\left(z - \left(-\sqrt{3} + i\right)\right)\left(z - (-2i)\right)$

$f(z) = z^3 - 8i$

or any equivalent form

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Question 2f.



z_3 is at the same perpendicular distance from the ray as (0, 1).

$a = 1 \times \cos 30^\circ, \quad |b| = 1 \times \sin 30^\circ$

$$z_3 = \frac{\sqrt{3}}{2} - \frac{i}{2}$$

Question 3a.

$$\frac{x}{15 + 15t}$$

Question 3b.

$$\frac{dx}{dt} = \text{rate in} - \text{rate out}$$

$$\frac{dx}{dt} = \frac{4}{(1+t)^2} \times 30 - \frac{x}{15+15t} \times 15$$

$$\frac{dx}{dt} = \frac{120}{(1+t)^2} - \frac{x}{1+t}$$

$$\frac{dx}{dt} + \frac{x}{1+t} = \frac{120}{(1+t)^2}$$

Convincing working leading to the result was required.

Question 3ci.

$$\frac{dx}{dt} = \frac{100}{(1+t)^2} - \frac{120 \log_e(1+t)}{(1+t)^2}$$

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Question 3cii.

$$\text{LHS} = \frac{100}{(1+t)^2} - \frac{120\log_e(1+t)}{(1+t)^2} + \frac{120\log_e(1+t)}{(1+t)^2} + \frac{20}{(1+t)^2}$$

$$= \frac{120}{(1+t)^2}$$

$$= \text{RHS}$$

Initial condition: $t = 0, x = \frac{120\log_e(1)}{(1)} + \frac{20}{(1)} = 20$

Question 3d.

Maximum rate = 30.8 when $t = 0.4$.

Question 4a.

$$m_1 g \sin\theta - k_1 m_1 g \cos\theta = 0$$

$$m_1 g \times \frac{3}{5} - k_1 m_1 g \times \frac{4}{5} = 0$$

$$k_1 = \frac{3}{4}$$

Question 4bi.

$$m_2 g \sin\phi - k_2 m_2 g \cos\phi = m_2 \times 0.2g$$

$$m_2 g \times \frac{2.8}{5.3} - k_2 m_2 g \frac{4.5}{5.3} = m_2 \times 0.2g$$

$$k_2 = \frac{29}{75}$$

Question 4bii.

$$v^2 - u^2 = 2as$$

$$(2v_2)^2 - {v_2}^2 = 2 \times 0.2g \times 5.3$$

$$v_2^2 = \frac{53}{75}g$$

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Question 4c.

$-k_3 m_1 g = m_1 a$

$a = -k_3 g$

$0^2 - v_1^{\,2} = 2 \times -k_3 g \times s_1$

$s_1 = \dfrac{v_1^{\,2}}{2g\,k_3}$

Convincing working leading to the result was required.

Question 4d.

$-k_3 m_2 g = m_2 a$

$a = -k_3 g$

$0^2 - (2v_1)^2 = 2 \times -k_3 g \times s_2$

$s_2 = \dfrac{2v_1^{\,2}}{k_3 g}$

$s_1 + s_2 < 2$

$\dfrac{v_1^{\,2}}{2g\,k_3} + \dfrac{2v_1^{\,2}}{k_3 g} < 2$

$v_1^2 < \dfrac{4k_3 g}{5}$

Question 5a.

$\bar{x} = 0.5, \;\; s = 0.0016$

Question 5b.

$H_0 : \mu = 0.5, \;\; H_1 : \mu > 0.5$

Question 5c.

$0.425 \; (0.426 \text{ also accepted})$

Question 5d.

$p > 0.05 \Rightarrow$ do not reject H_0

Question 5e.

0.5026

Question 5f.

$a = 0.496, \;\; b = 0.504$

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Question 5g.

$${}^5C_1 \times 0.02^1 \times 0.98^4 = 0.092$$

Question 6a.

$$y = 500 - \frac{x}{30}$$

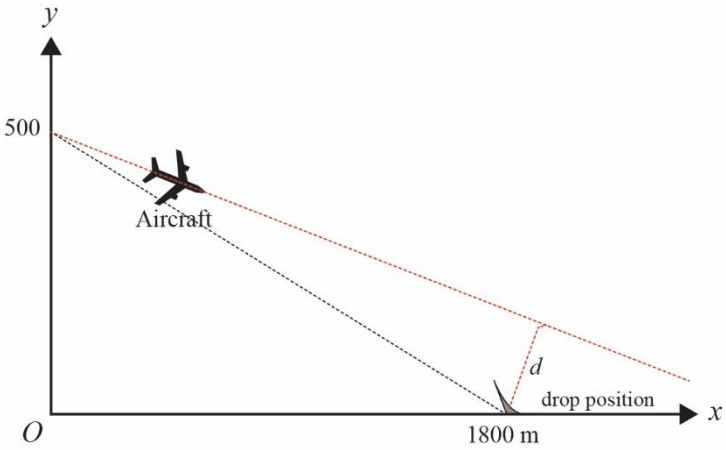
Question 6b.

$$\dot{\underset{\sim}{r}}(t) = 60\underset{\sim}{i} - 2\underset{\sim}{j}$$

$$\left|\dot{\underset{\sim}{r}}(t)\right| = \sqrt{60^2 + 2^2}$$

$$= 60.03$$

Question 6c.



$$d = \left(-1800\underset{\sim}{i} + 500\underset{\sim}{j}\right) \cdot \frac{\left(2\underset{\sim}{i} + 60\underset{\sim}{j}\right)}{\sqrt{3604}} = 439.76$$

Alternatively, minimise $d = \sqrt{(60t - 1800)^2 + (500 - 2t)^2}$.

Question 6d.

$$460 = 2t + \frac{1}{2} \times 9.8 \times t^2$$

$$t = 9.4871\text{s}$$

$$\text{Horizontal distance} = 9.4871 \times 60$$

$$460 = 500 - \frac{x}{30} \Rightarrow x = 1200$$

$$\text{Shortfall} = 1800 - (1200 + 9.4871 \times 60) = 30.8\,\text{m}$$

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Question 6e.

$$h = 2t + \frac{1}{2} \times 9.8 \times t^2$$

$$t = \frac{-2 + \sqrt{4 + 19.6h}}{9.8}$$

$$h = 500 - \frac{x}{30} \Rightarrow x = 30(500 - h)$$

$$30(500 - h) + 60 \times \left(\frac{-2 + \sqrt{4 + 19.6h}}{9.8} \right) = 1800$$

$$h = 458.95\,\text{m}$$

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